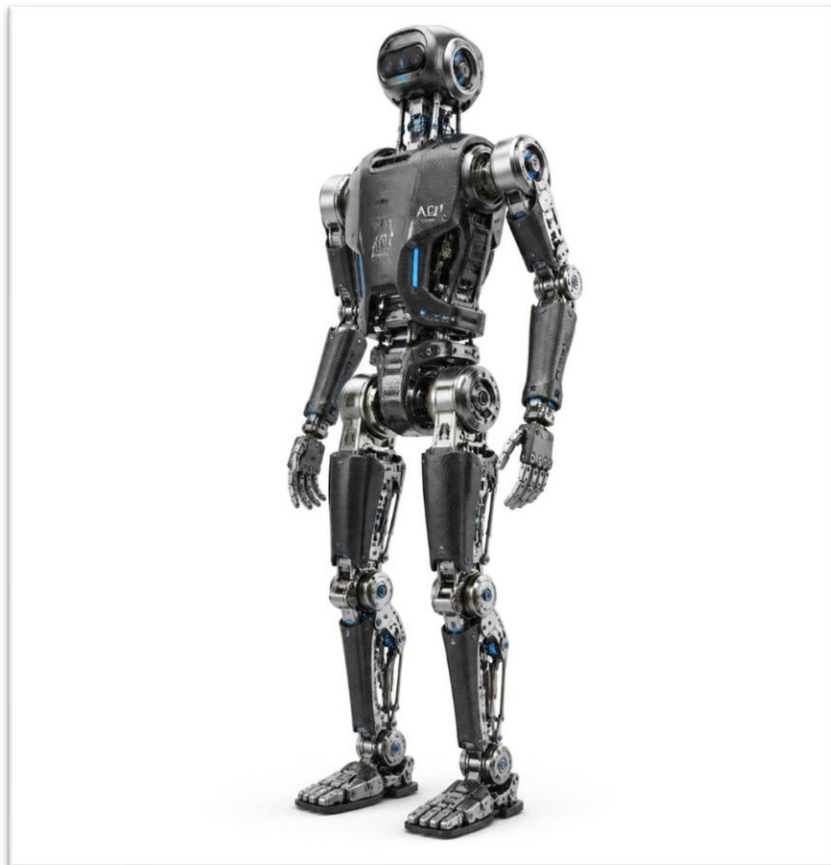


# HUMANOID ROBOT DESIGN CONCEPTS

## *Design Concepts and Architecture of a Humanoid Robotic System*

|                         |                                                |
|-------------------------|------------------------------------------------|
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## 1. Executive Summary

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Approximately 80% of global industrial infrastructure is built around human ergonomics and reach envelopes. As automation demand extends into spaces designed natively for human workers — stairs, narrow aisles, and equipment sized for human hands — wheeled mobile robots and fixed-base manipulators face structural limitations that a human-scale, bipedal form factor can address.

This document presents the preliminary design concept and system architecture for a modular, bipedal Humanoid Robotic Platform intended for operation in human-centric industrial, inspection, and disaster-response environments. The proposed architecture integrates a 28-Degree-of-Freedom (DoF) mechanical structure, a hybrid Quasi-Direct Drive (QDD) and Harmonic Drive actuation strategy, and a multi-tiered control stack built on ROS 2 middleware.

Dynamic balance is addressed through a Model Predictive Control (MPC) and Whole-Body Control (WBC) hierarchy informed by Zero Moment Point (ZMP) and Center of Mass (CoM) estimation. Electrical power is distributed through a 48 V DC bus managed by a custom Power Distribution Unit (PDU) and Battery Management System (BMS). Verification is planned through a staged simulation pipeline (Gazebo and NVIDIA Isaac Sim) prior to Hardware-in-the-Loop (HIL) and physical prototype testing over a 24-week development roadmap with a preliminary budget of \$35,000.

## 2. Problem Statement

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### 2.1 Operational Challenge

Most conventional industrial automation relies on specialized environments:

- **Wheeled Mobile Robots (AGVs/AMRs):** restricted to flat, obstruction-free surfaces and incapable of vertical traversal (e.g., staircases, ladders).
- **Static Manipulators:** highly efficient within fixed work cells but lacking the mobility to transition between disparate tasks or locations.

Human environments — factories, warehouses, and disaster sites — are optimized for human morphology. Deploying automation into these spaces requires a form factor capable of overcoming structural barriers designed around human feet, hands, and sensory reach.

## 3. Project Objectives & Scope

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- Define a kinematic architecture reproducing human reach and stride envelopes without unnecessary redundancy.
- Select an actuation strategy balancing impact tolerance, backdrivability, and payload precision.
- Establish a hierarchical control architecture for real-time dynamic balance and whole-body motion.
- Define a power architecture sustaining peak torque demand with thermal/electrical protection.
- Establish a staged simulation-to-hardware verification methodology prior to physical prototyping.

Scope: kinematic allocation across torso, limbs, and head; actuation, sensing, and power-management selection; control loops for locomotion and balance; and a simulation pipeline for pre-hardware validation. Design constraints: human-comparable form factor, 48 V DC bus, 1 kHz joint-control loop over EtherCAT, and off-the-shelf QDD / Harmonic Drive actuator classes (no custom actuator development assumed).

## 4. System Architecture

The robot is structured into three subsystems coordinated over a centralized ROS 2 bus (Figure 1): (a) Mechanical — an articulated aluminum-carbon exoskeleton housing 28 active joints; (b) Electrical & Power — a 48 V DC primary bus stepped down via a custom PDU, with an integrated BMS; and (c) Software & Control — a real-time 1 kHz joint-control tier plus a high-level 30–60 Hz perception/planning tier, both connected through ROS 2 middleware. Perception, inertial, and force-torque data are fused at a central compute node driving the MPC/WBC hierarchy, which issues torque commands to actuator drivers over EtherCAT.

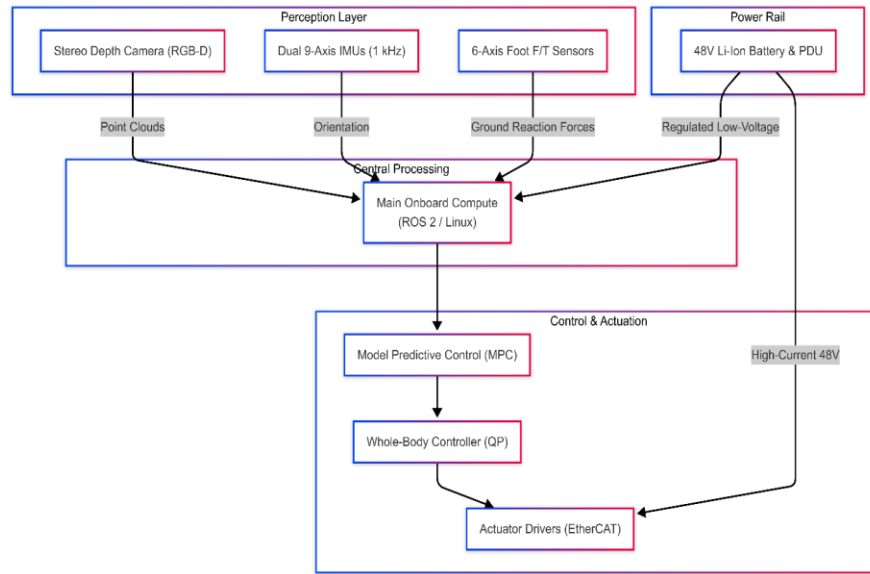


Figure 1. Overall System Architecture — Perception, Power, and Control Integration.

## 5. 28-DoF Kinematic Configuration

28 active DoF are distributed across five body segments (Table 1), optimized to match human reach and stride characteristics while avoiding redundancy.

| Body Segment          | Joint Configuration                | DoF | Function                                       |
|-----------------------|------------------------------------|-----|------------------------------------------------|
| Lower Limbs (×2 legs) | Hip (3), Knee (1), Ankle (2)       | 12  | ZMP-aware stance, gait & lateral stabilization |
| Torso / Waist         | Waist assembly (Yaw, Pitch)        | 2   | CoM shifting during gait, upper-body rotation  |
| Upper Limbs (×2 arms) | Shoulder (3), Elbow (1), Wrist (2) | 12  | Payload handling & end-effector orientation    |
| Head                  | Pan/Tilt neck                      | 2   | Visual tracking, sensor alignment              |
| Total                 |                                    | 28  | —                                              |

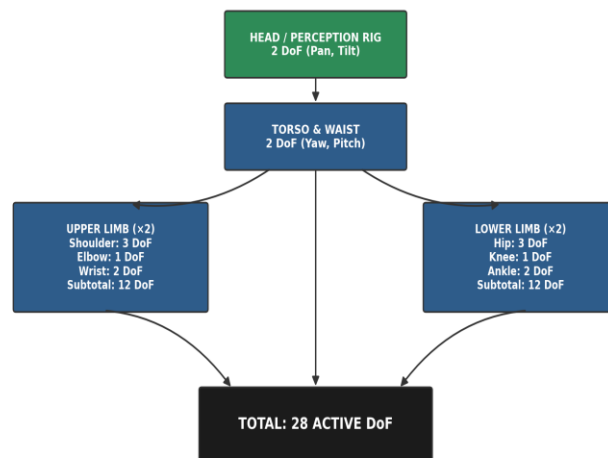


Figure 2. 28-DoF Kinematic Configuration and Joint Allocation by Body Segment.

## 6. Actuation Strategy

| Parameter     | Lower Body (Hip/Knee)                    | Upper Body (Shoulder/Wrist)              |
|---------------|------------------------------------------|------------------------------------------|
| Actuator Type | Quasi-Direct Drive (QDD)                 | Harmonic Drive Assembly                  |
| Gear Ratio    | 6:1 – 10:1                               | 50:1 – 100:1                             |
| Advantage     | High backdrivability, impact resistance  | High torque density, near-zero backlash  |
| Application   | Dynamic walking, jump-landing absorption | Precise trajectory & payload positioning |
| Feedback      | Dual absolute magnetic encoders          | High-resolution optical shaft encoders   |

Lower-limb QDD actuators use low gear ratios, giving high backdrivability so ground-reaction forces transfer back to the motor coils; impacts are absorbed electromechanically rather than by gear teeth, enabling active impedance control for compliant locomotion.

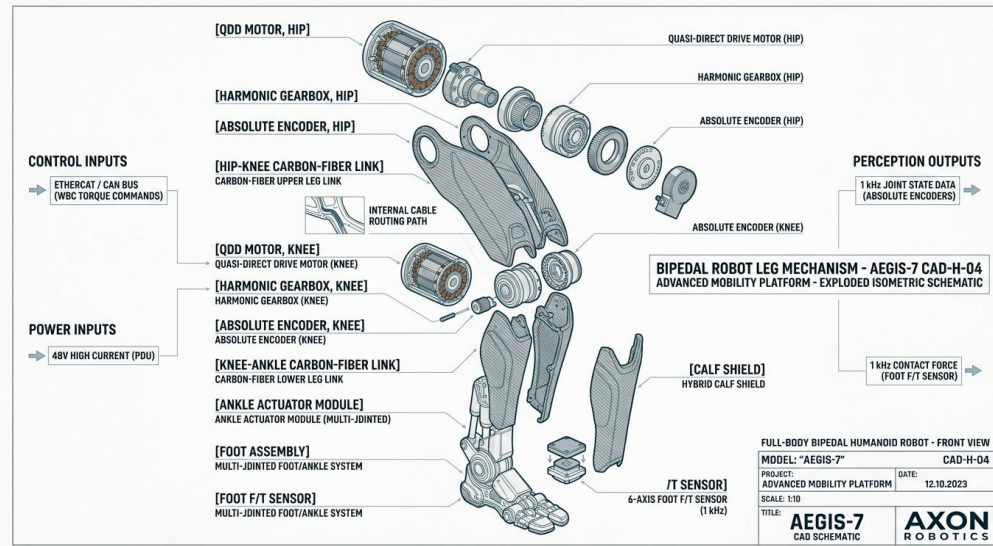


Figure 3. Representative Lower-Limb Actuator Assembly (generic reference illustration carrying third-party branding — "AXON Robotics," model "AEGIS-7" — unrelated to this project; not the platform's own CAD output).

## 7. Sensors

- Joint encoders — dual absolute magnetic (QDD) / optical shaft (Harmonic Drive), closing the 1 kHz EtherCAT loop.
- IMU — dual 9-axis, 1 kHz, for orientation and CoM tracking.
- Foot force-torque sensors (×2, 6-axis) — ground reaction forces for ZMP estimation.
- RGB-D / stereo perception — head-mounted rig for spatial mapping and object recognition.
- Tactile sensor arrays — end-effector contact sensing (budgeted; placement pending hand design).

## 8. Electrical & Power Architecture

Power is distributed from a 48 V DC primary bus through a custom PDU that regulates high-current motor rails and steps down 5 V/12 V logic supplies (Table 2). Two 48 V, 15 Ah LiFePO4 packs supply peak actuator current; a BMS monitors cell voltage, temperature, and discharge rate. Safety disconnect relays and TVS diode suppression provide fault-triggered and transient protection.

| Subsystem               | Voltage                       | Function                              | Protection                            |
|-------------------------|-------------------------------|---------------------------------------|---------------------------------------|
| Battery Packs (×2)      | 48 V, 15 Ah                   | Primary energy storage (LiFePO4)      | BMS thermal/discharge monitoring      |
| Power Distribution Unit | 48 V in / 5 V, 12 V, 48 V out | Regulates logic + motor-drive rails   | TVS suppression, decoupling caps      |
| Motor Drive Rail        | 48 V, high current            | Powers QDD/Harmonic drivers           | Overcurrent protection, E-stop relays |
| Logic Rail              | 5 V / 12 V                    | Compute, sensors, control electronics | Regulated low-voltage output          |

## 9. Control Architecture

Sensors → State Estimation → Planning → MPC → Whole-Body Control → Joint Controllers → EtherCAT → Motor Drivers → Actuators → Robot (Figure 4). ROS 2 hosts the high-level tier (30–60 Hz: perception, mapping, trajectory synthesis), decoupled from the hard-real-time joint-control loop (1 kHz). MPC generates dynamically feasible CoM/footstep trajectories over a receding horizon; WBC resolves these into joint-space torque commands across all 28 DoF via QP optimization, respecting joint, torque, and contact constraints. ZMP, derived from foot F/T sensing and CoM tracking, bounds the WBC's admissible support region. Each joint closes a local position/velocity/torque loop using encoder feedback, with EtherCAT providing the deterministic, low-latency fieldbus for the 1 kHz control cycle.

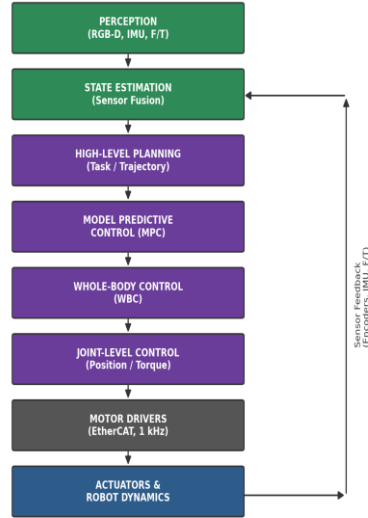


Figure 4. Hierarchical Control Loop: Perception to Actuation with Sensor Feedback.

## 10. Simulation & Virtual Verification

URDF → Gazebo / Isaac Sim → HIL → Hardware Validation (Figure 5). The robot's structure is modeled in URDF; Gazebo simulates rigid-body dynamics, ZMP, and foot-ground contact under varied gravity/friction; NVIDIA Isaac Sim generates synthetic RGB-D data to stress-test perception ahead of hardware availability; HIL testing verifies real-time loop stability at 1 kHz before deployment to physical motor drivers.

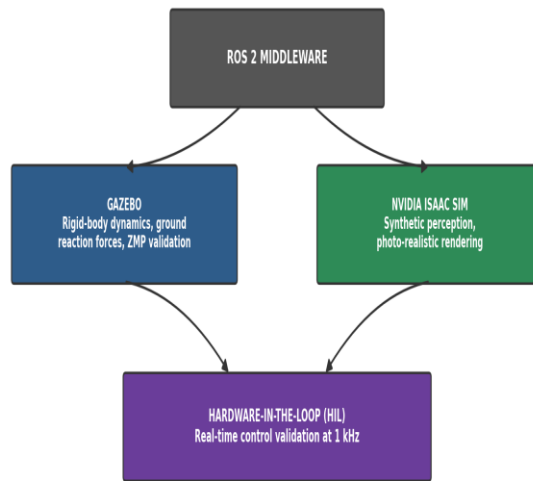


Figure 5. ROS 2 → Gazebo / Isaac Sim → HIL Verification Pipeline.

## 11. 24-Week Project Roadmap

| Phase | Weeks | Milestone              | Deliverables                                                     |
|-------|-------|------------------------|------------------------------------------------------------------|
| 1     | 1–4   | CAD & URDF             | 3D assembly models, inertial properties, structural FEA          |
| 2     | 5–8   | Simulation             | Gazebo/Isaac Sim setup, ROS 2 control package, MPC tuning        |
| 3     | 9–12  | Hardware Procurement   | AI 7075 machining, motor-driver PCB, PDU assembly                |
| 4     | 13–16 | Subsystem Integration  | Single-limb bench test, EtherCAT setup, encoder calibration      |
| 5     | 17–20 | Gait & Locomotion      | Flat-ground bipedal walking, foot F/T integration, fall-recovery |
| 6     | 21–24 | Full-System Validation | Obstacle traversal, arm-leg coordination, full demo              |

## 12. Target Applications

- Industrial Logistics & Warehousing — payload transport across uneven facilities, stairs, legacy infrastructure.
- Disaster Response & Search/Rescue — debris traversal, latch/valve operation, human power-tool use.
- Facility Inspection & Maintenance — narrow catwalks, gauge reading, infrastructure inspection.

## 13. Risk Assessment

| Risk                                         | Severity | Probability | Mitigation                                                               |
|----------------------------------------------|----------|-------------|--------------------------------------------------------------------------|
| Thermal — actuator winding overheating       | High     | Medium      | Dynamic thermal throttling; active cooling channels in hip/knee housings |
| Dynamic Stability — MPC divergence / falls   | High     | Medium      | Virtual fall-recovery boundaries; passive-compliance fallback modes      |
| Power — voltage drops / current spikes       | Medium   | High        | High-discharge LiFePO4 cells; PDU with TVS suppression                   |
| Communication — EtherCAT packet drops/jitter | Medium   | Low         | PREEMPT_RT kernel pinned to dedicated cores for motor-bus loop           |

## 14. Preliminary Bill of Materials

Preliminary Engineering Estimate — costs subject to final supplier selection and specification.

| Category        | Component                                   | Qty   | Est. Total |
|-----------------|---------------------------------------------|-------|------------|
| Actuation       | Lower-Limb QDD Actuators                    | 12    | \$10,800   |
| Actuation       | Upper-Body Harmonic Drive Actuators         | 14    | \$5,600    |
| Actuation       | Neck Pan/Tilt Actuators                     | 2     | \$600      |
| Actuation       | Dual Absolute Magnetic Encoders             | —     | \$1,200    |
| Compute/Sensing | Onboard Compute (Jetson AGX Orin / Ind. PC) | 1     | \$2,200    |
| Compute/Sensing | 6-Axis Foot Force-Torque Sensors            | 2     | \$3,000    |
| Compute/Sensing | Head Stereo RGB-D Rig                       | 1     | \$850      |
| Compute/Sensing | Dual 9-Axis IMUs                            | 2     | \$1,000    |
| Compute/Sensing | Tactile Sensor Arrays & Cabling             | —     | \$1,000    |
| Structural      | CNC Al 7075-T6 Internal Skeleton            | 1 set | \$3,200    |
| Structural      | CFRP Shell Fairings & Enclosures            | 1 set | \$1,700    |
| Power           | 48 V 15 Ah Battery Packs                    | 2     | \$1,600    |
| Power           | Custom PDU & BMS PCBs                       | 1 set | \$1,200    |
| Power           | Cabling, Connectors, Safety Relays          | —     | \$1,050    |

**TOTAL ESTIMATED PROTOTYPE COST: \$35,000 (Actuation \$18,200 · Compute/Sensing \$8,050 · Structural \$4,900 · Power \$3,850)**

## 15. Conclusion

This proposal lays out the structural blueprint for a modular 28-DoF humanoid robot concept for unstructured, human-designed environments. Pairing lower-limb QDD actuation with a hierarchical MPC + WBC control structure, the design reconciles dynamic agility with impact resistance, while a ROS 2 control loop couples high-rate state estimation with low-latency torque generation. A risk-mitigated roadmap bridges Gazebo/Isaac Sim simulation to physical hardware validation over 24 weeks, targeting industrial logistics, disaster response, and facility-inspection applications.

## 16. Future Work

- Reinforcement-learning locomotion — transitioning from classical MPC toward end-to-end RL policies for terrain adaptation.
- Advanced autonomous manipulation — integrating dexterous grasping and force-controlled manipulation pipelines.
- Expanded perception / VLA capabilities — vision-language-action models for voice-commanded task execution.

## 17. References

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